MSG No. 59.414 P_Bmmn [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[1,6,10,13]`
2	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[4,7,11,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,10,13]`
2	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[4,7,11,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[2,3,5,8]`
3	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[9,12,14,15]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[10,11,13,16]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,5]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,6]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,7]`
4	$[0, 0, 0]$	`[4,8]`
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,13]`
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[10,14]`
7	$\left[0, 0, \frac{1}{2}\right]$	`[11,15]`
8	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[12,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	`[1,10]`
2	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[2,9]`
3	$\left[-x, 0, \frac{1}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[4,11]`

continued ...

Table 5

No.	position	mapping
5	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[5,14]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[6,13]
7	$\left[x,\ 0,\ \frac{3}{4}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[8,15]

Table 6: Wyckoff site: 8f, site symmetry: m..

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	[1,6]
2	$\left[\frac{3}{4},\ -y,\ -z\right]$	[2,5]
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	[3,8]
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	[4,7]
5	$\left[\frac{3}{4},\ y,\ z+\frac{1}{2}\right]$	[9,14]
6	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}-z\right]$	[10,13]
7	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11,16]
8	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12,15]

Table 7: Wyckoff site: 8g, site symmetry: .m.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	[1,7]
2	$\left[x+\frac{1}{2},\ \frac{3}{4},\ -z\right]$	[2,8]
3	$\left[-x,\ \frac{3}{4},\ -z\right]$	[3,5]
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z\right]$	[4,6]
5	$\left[x+\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[9,15]
6	$\left[x,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[10,16]
7	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[11,13]
8	$\left[-x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[12,14]

Table 8: Wyckoff site: 16h, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	[2]
3	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[\frac{1}{2}-x,\ y,\ z\right]$	[6]
7	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[7]
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[9]
10	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11]

continued ...

Table 8

No.	position	mapping
12	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[16]